

Pre-Hackathon Consultation — Agenda

Project: Spatial Degradation of Hepatocyte Zonation **Team:** Roei & Shira **Course:** Comp. Genomics 76553

One line: use Paper 2's healthy human-liver zonation atlas as a quantitative “ruler” to give Paper 1's spatially-blind disease hepatocytes a pseudo-zonation coordinate, then measure how zonation collapses across MASLD stages (H1), which gene programs collapse (H2), and whether de-zonated cells turn plastic (H3).

1. Methodology (the calls that could change direction)

1. **Transcriptome-wide vs. landmark signature.** Our stance: the *full* ~1,600-gene program is the primary result; the canonical 13+8 landmark set is a reported *audit*; the validation battery picks which leads. Is this appropriately novel without over-reaching?
2. **Cross-dataset ruler-validity (our central risk).** We apply a *healthy* Paper 2 ruler to *disease* Paper 1 cells across platforms. What would convince you the transfer is valid? Concerns about batch/platform confounds beyond our coherence / anti-correlation / split-half checks?
3. **Classifier ground-truth labels.** We currently use a tercile-of-coordinate *fallback* label. Paper 2 has Visium-mapped zonation that could give *true* labels — worth porting for a defensible classifier, or is the fallback acceptable?
4. **H2 circularity.** The coordinate is built from genes, then we ask which genes lose zonation. We guard with a repeated held-out gene split. Does that satisfy you, or do you prefer the per-gene interaction model ($\text{expression} \sim \text{coord} + \text{stage} + \text{coord} \times \text{stage}$)?

2. Scope & expectations

1. **Core deliverable for two days.** Would you rather see **H1** done rigorously, or **H1–H3** done lightly? Where should we spend the time?
2. **Small n at the extremes.** Cirrhosis ($n=4$) and end-stage ($n=5$) rest on few donors. Acceptable as a stated limitation with donor-level confidence intervals, or a problem you'd want handled differently?